



Basic Principles of Medicine 2  
Module: Endocrine & Reproduction | DLA

# Histology of the Male Reproductive System

**Dr. Kevlian Andrew**

---

*[kandrew4@sgu.edu](mailto:kandrew4@sgu.edu)*

*Department of Anatomical Sciences  
School of Medicine, St. George's University*

# Copyright

All year-1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic, or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

'View only' files, such as lecture recordings, are explicitly excluded from download, and creating copies of these recordings by students and other users is strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St. George's University. Users of this document identifying potential violations of these regulations are asked to bring their concerns to the attention of the author.



# Recommended Reading

- Histology – A Text and Atlas Pawlina 9<sup>th</sup> Edition Chapter 22: Male Reproductive System

<https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290>

## Other Resources

- Mescher, A. L. (2018). The Male Reproductive System. In *Junqueira's Basic Histology Text & Atlas* (15th ed., pp. 439–459). McGraw Hill Education.

# Objectives

|                               |                                                                                                             |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| SOM.MK.I.BPM2.2.ER.1.HCB.1560 | Describe the intratesticular tubular path travelled by spermatozoa.                                         |
| SOM.MK.I.BPM2.2.ER.1.HCB.1561 | Describe the extratesticular tubular path travelled by spermatozoa.                                         |
| SOM.MK.I.BPM2.2.ER.1.HCB.1562 | Give the function(s) of the tubuli recti.                                                                   |
| SOM.MK.I.BPM2.2.ER.1.HCB.1563 | Describe the microscopic structure of the tubuli recti.                                                     |
| SOM.MK.I.BPM2.2.ER.1.HCB.1564 | Give the function(s) of the rete testis.                                                                    |
| SOM.MK.I.BPM2.2.ER.1.HCB.1565 | Describe the microscopic structure rete testis.                                                             |
| SOM.MK.I.BPM2.2.ER.1.HCB.1566 | Give the function(s) of the efferent ductules.                                                              |
| SOM.MK.I.BPM2.2.ER.1.HCB.1567 | Describe the microscopic structure of the efferent ductules.                                                |
| SOM.MK.I.BPM2.2.ER.1.HCB.1568 | Give the function(s) of the epididymis.                                                                     |
| SOM.MK.I.BPM2.2.ER.1.HCB.1569 | Describe the microscopic structure of the epididymis.                                                       |
| SOM.MK.I.BPM2.2.ER.1.HCB.1570 | State the function of the ductus deferens.                                                                  |
| SOM.MK.I.BPM2.2.ER.1.HCB.1571 | Describe the histological appearance of the structure of the ductus deferens.                               |
| SOM.MK.I.BPM2.2.ER.1.HCB.1572 | State the function of the erectile bodies of the penis.                                                     |
| SOM.MK.I.BPM2.2.ER.1.HCB.1573 | Describe the histological appearance of the structure of the erectile bodies of the penis.                  |
| SOM.MK.I.BPM2.2.ER.1.HCB.1574 | <b>Describe the characteristics and composition of semen.</b>                                               |
| SOM.MK.I.BPM2.2.ER.1.HCB.1575 | Distinguish histologically between the structure of the prostatic, membranous and penile (spongy) urethras. |
| Covered in Anatomy            | Describe the blood supply to the erectile tissue of the penis.                                              |
| Covered in Anatomy            | Compare and contrast erection and ejaculation.                                                              |



# Learning Outcomes

By the end of this DLA, students should be able to:

- **Describe the path of spermatozoa from the seminiferous tubules to the external urethral orifice**
- **List, identify and describe the structure and function of all structures that form this path**
- **Describe the histology of the penis**

Please contact Dr. Andrew with any questions:

[kandrew4@sgu.edu](mailto:kandrew4@sgu.edu)

# Male Reproductive System - Overview

## 1. Testes

- Seminiferous tubules
- Intratesticular ducts

## 2. Excurrent Ducts

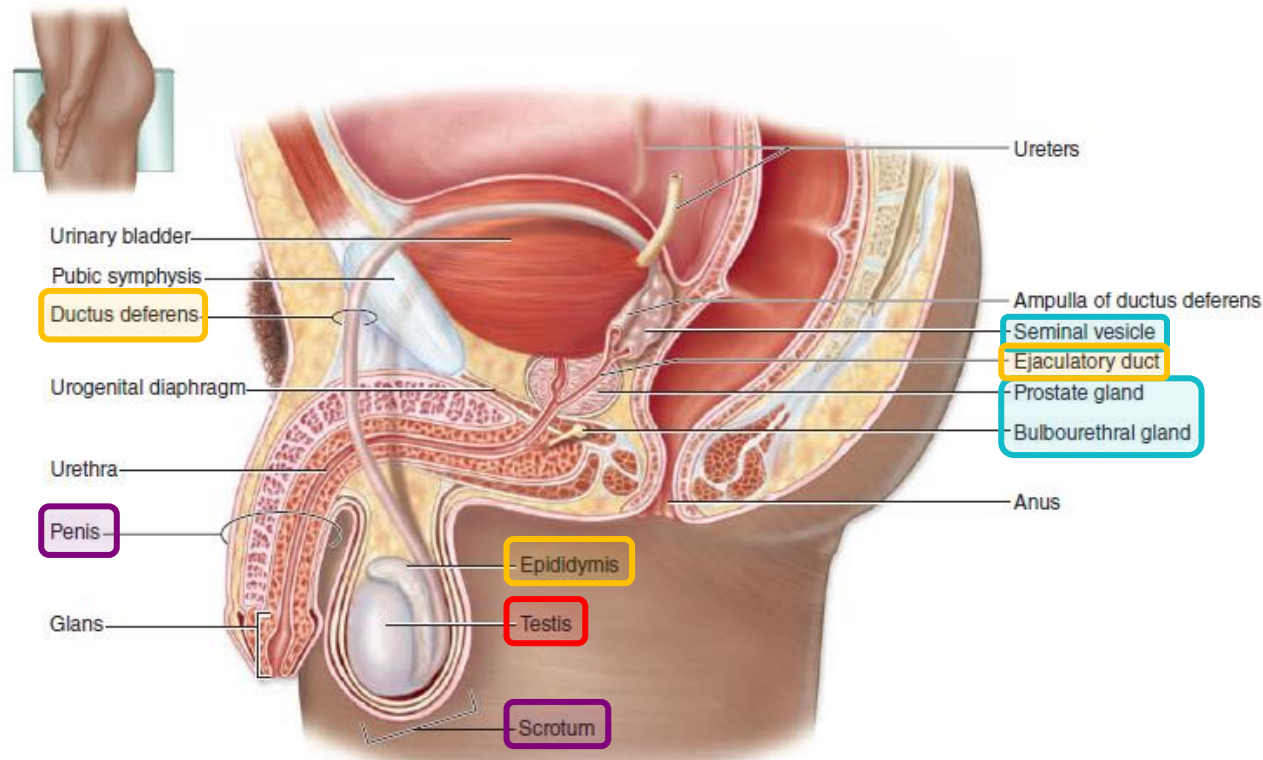
- Efferent tubules
- Epididymis
- Ductus deferens
- Ejaculatory duct

## 3. Accessory Glands

- Seminal vesicles
- Prostate gland
- Bulbourethral glands

## 4. External Genitalia

- Penis and urethra
- Scrotum



# Tubular Pathway of Spermatozoa

## 1. Seminiferous tubules

- Convoluted part  
→ Spermatogenesis

## 2. Straight tubules

- straight terminal part of seminiferous tubules

## 3. Rete testis

## 4. Efferent ductules

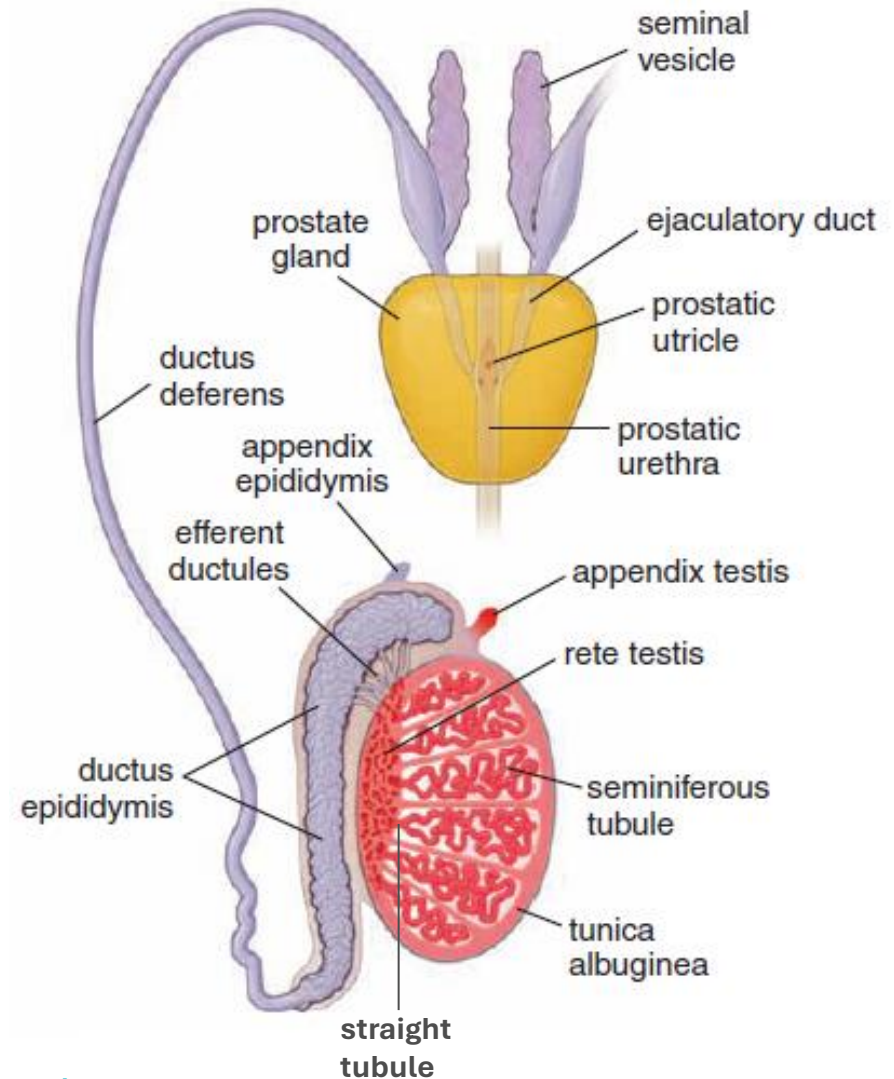
## 5. Duct of epididymis

## 6. Ductus (vas) deferens

## 7. Ejaculatory duct

## 8. Urethra (see later)

- *Intratesticular ducts (genital cords)*
- *Extratesticular ducts (mesonephric duct and tubules)*

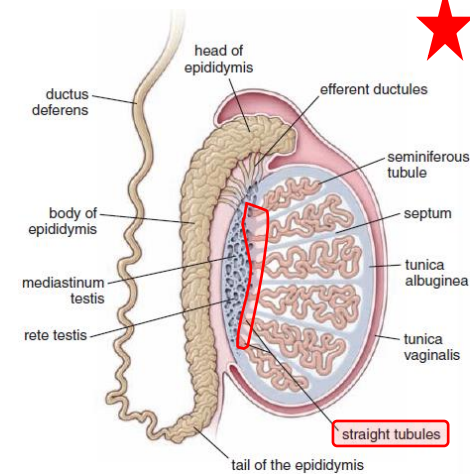




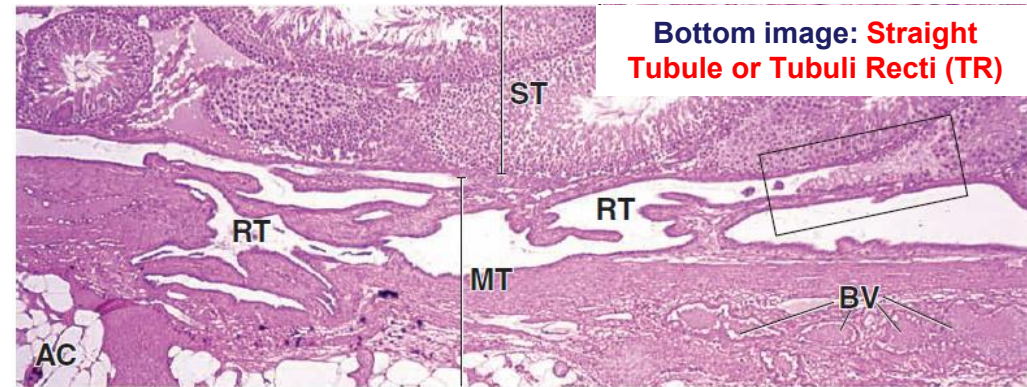


# Tubuli recti (Straight tubules)

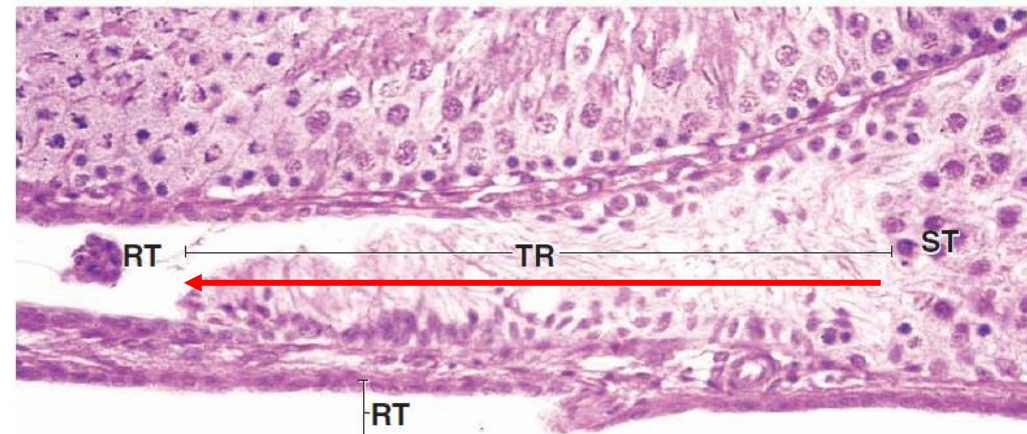
- Short terminal portion of seminiferous tubule (ST)
- **Lined only by Sertoli cells** (see lecture)
- Narrows near **distal end**
  - **Lined by simple cuboidal epithelium**
- Empties into Rete testis (RT)



- ✓ *Straight tubule starts on right at (ST) and ends on left at (RT) (see arrow)*
- ✓ *Epithelium becomes cuboidal toward end of straight tubule (TR)*
- ✓ *Transport sperm from ST to RT*



**Bottom image: Straight Tubule or Tubuli Recti (TR)**

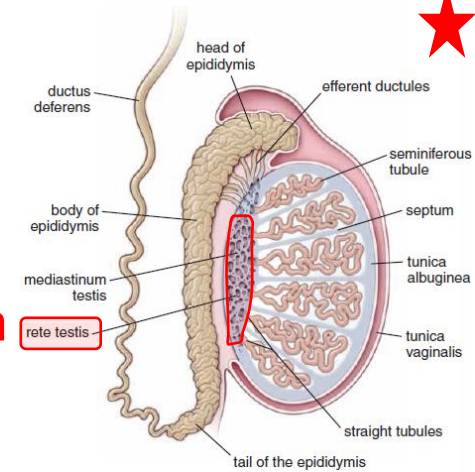






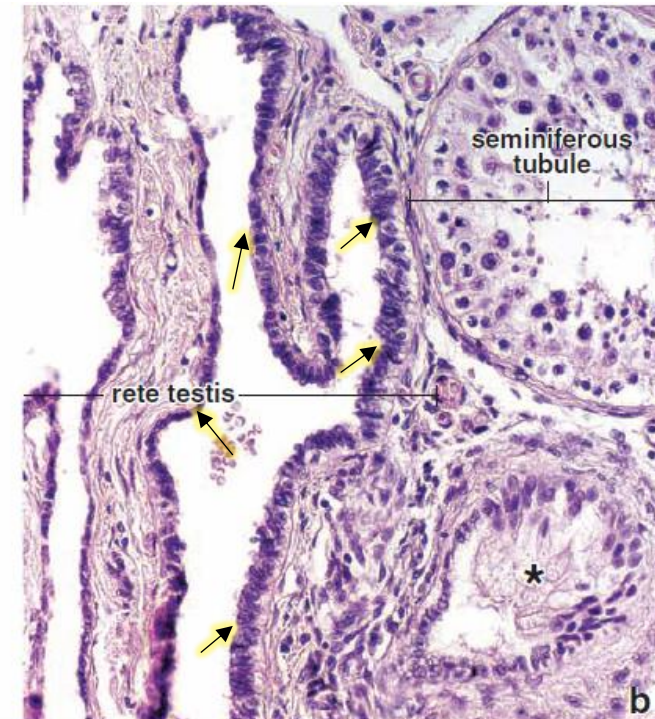
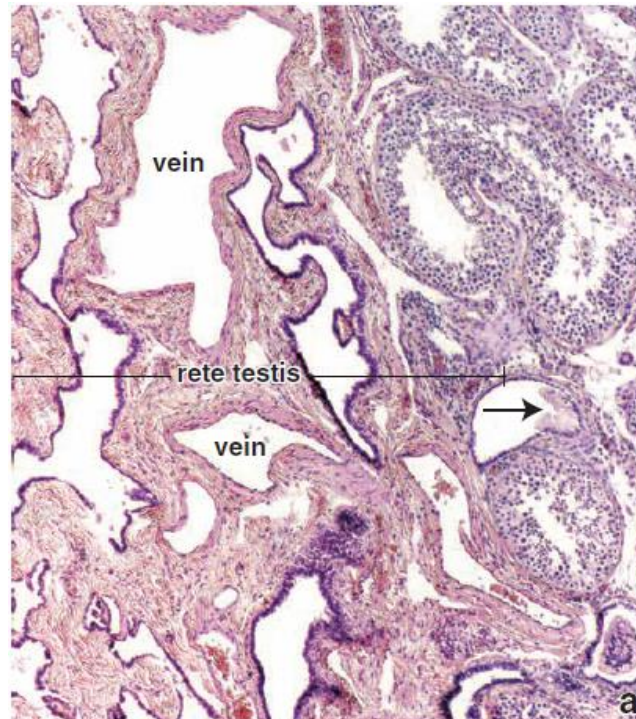
# Rete Testis

- Complex series of interconnecting channels
- Found in mediastinum testis (see lecture)
- Lined by **simple cuboidal** (low columnar) **epithelium**
- Apical specializations
  - **Single cilium**: transport sperm and fluid
  - **Relatively few, short microvilli**: absorb fluid



**(Image a) Arrow →:**  
termination of straight tubule (lined by Sertoli cells) → contents enter rete testis

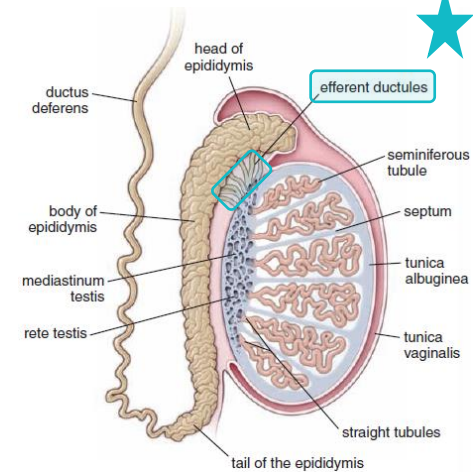
**(Image b) Asterisk\*:**  
terminating straight tubule entering the rete testis. Simple cuboidal/low columnar epithelium lining channel (→).





# Efferent Ductules

- ~12 to 20 ductules
- Become highly coiled as exits testis (*Coni vasculosi*)
  - 6–10 conical masses of highly convoluted ducts
  - Open into single channel at base
    - → Duct of epididymis
    - **Forms part of head of epididymis**
- Forms channel between rete testis and duct of epididymis
- **Functions:**
  - **Absorb most of fluid** secreted by seminiferous tubules
  - **Transport sperm**
    - Fibromuscular contraction
    - Ciliary action







# Efferent Ductules

- Lined by **pseudostratified columnar epithelium**
- Alternating groups of tall and short cells
  - Tall ciliated columnar cells**
    - Move contents toward epididymis
  - Basal cells** among columnar cells
    - Stem cells
  - Short non-ciliated cuboidal cells with numerous microvilli**
    - Absorb fluid from lumen
- Lumen: uneven ('saw-tooth') appearance**
- Several cell layers of **circularly arranged smooth muscle** surrounding ductules

**SM:** smooth muscle

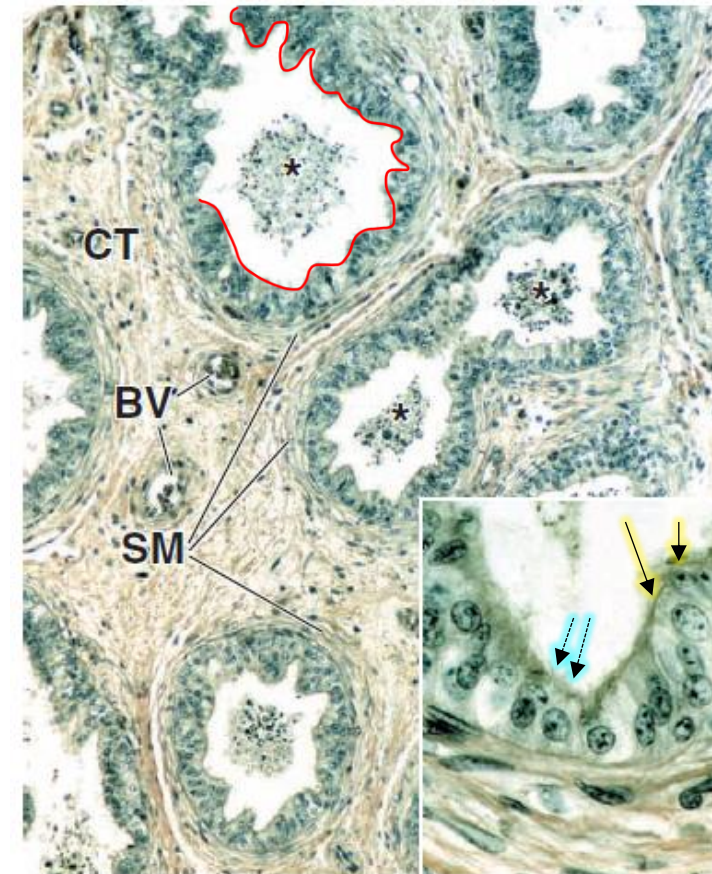
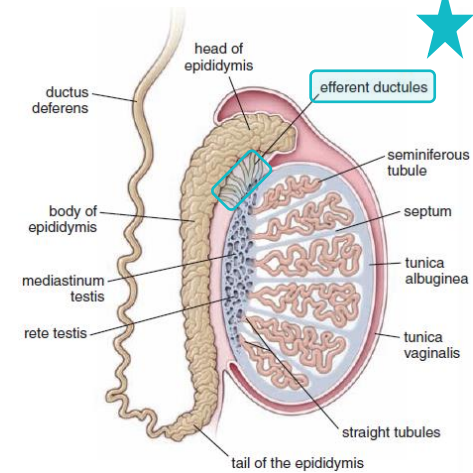
**CT:** connective tissue

**BV:** Blood vessel

**Asterisk \*:** clumps of spermatozoa in lumen

**Yellow arrow:** tall ciliated columnar cells (hills)

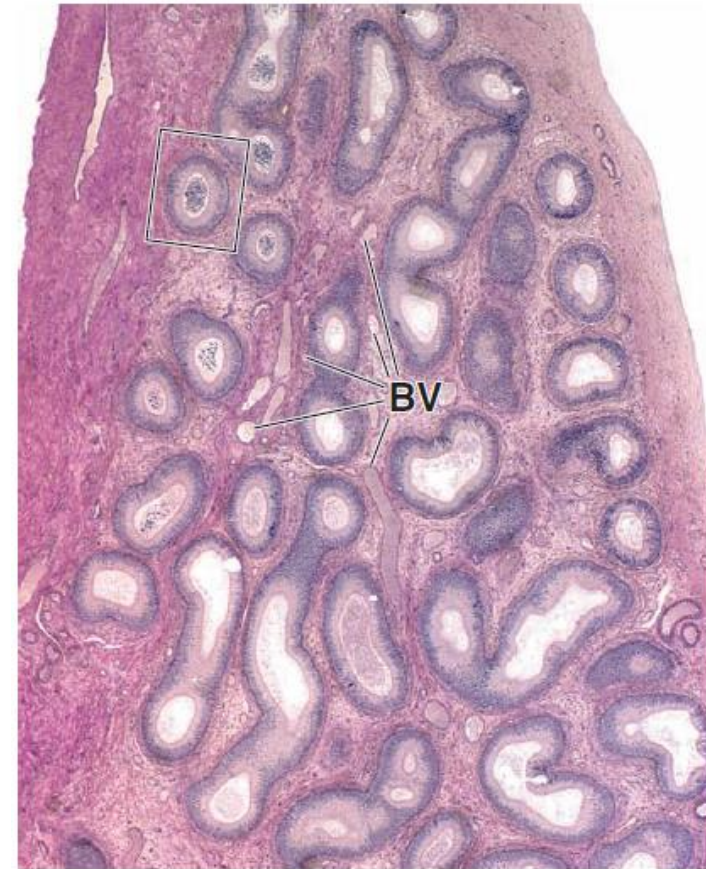
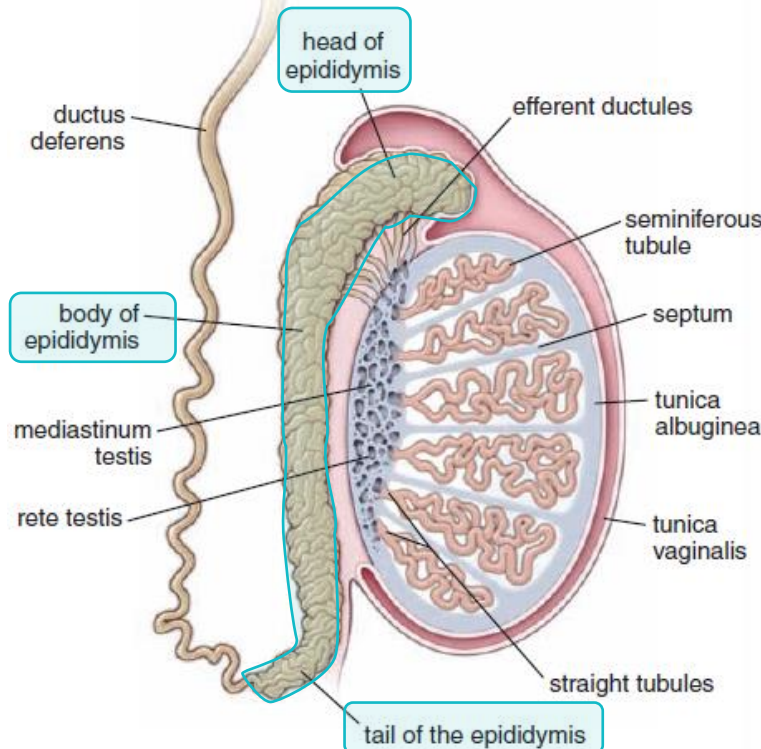
**Blue arrows:** cuboidal cells with microvilli (valleys)





# Epididymis

- Crescent-shaped structure, ~7.5cm long
- Lies along superior and posterior testes surface
- **Highly coiled tube (stretched length ~4-6 m)**
- Consists of **head, body, tail**
- Contains 2 types of ducts
  - **Efferent ductules (head)**
  - **Duct of epididymis (body and tail)**



*Varying profiles of ducts due to highly coiled nature (most coiled at head, less in body and tail)*



# Epididymis

## Lined by pseudostratified columnar epithelium

### 1. Principal cells (PC)

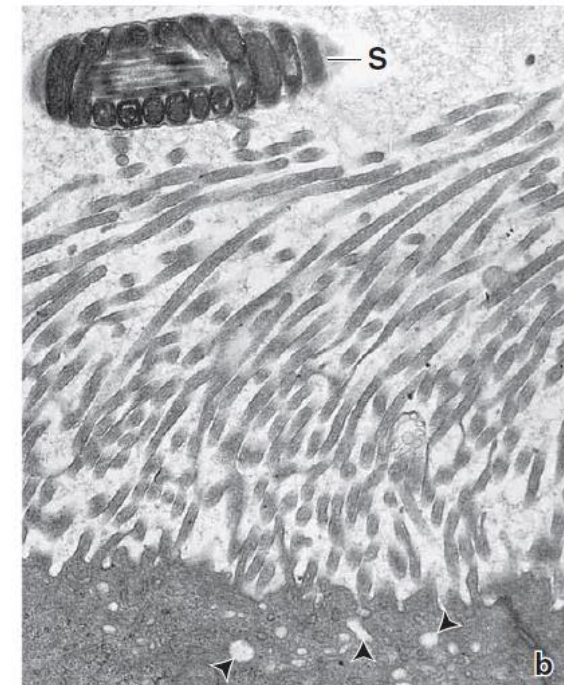
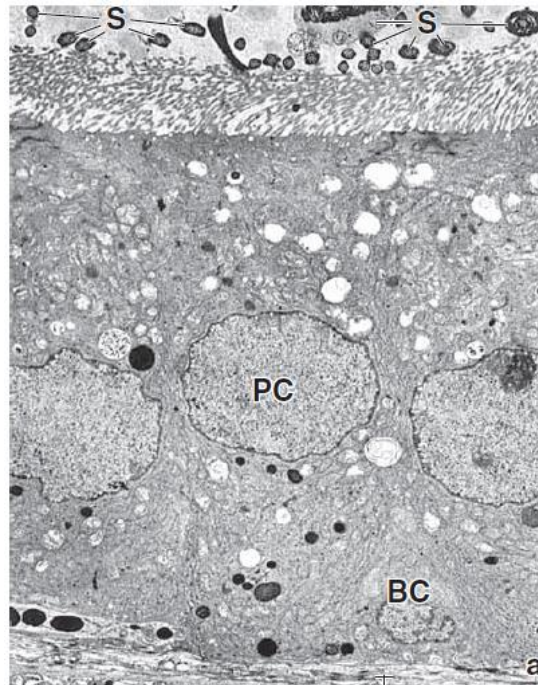
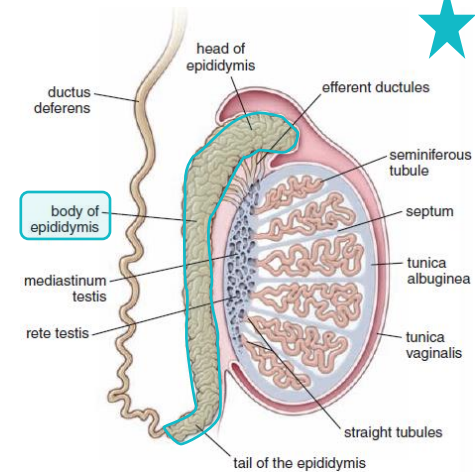
- Columnar cells with cylindrical nuclei
- **Stereocilia** - long modified **microvilli**
- Endocytotic vesicles (arrowheads), invaginations (base of stereocilia)
- **rER**, large supranuclear **golgi apparatus**
  - Secrete **glycerophosphocholine**, **sialic acid**, **glycoproteins**, **glycocalyx**, **steroids**
  - → **sperm (S)** maturation along passage

### 2. Basal cells (BC)

- Small round cells with spherical nuclei
- Stem cells found on basal lamina

### ❖ Halo cells

- Migrating lymphocytes
- Most proximal location of immune cells (normally – **Why?**)





# Epididymis

## Smooth muscle coat

- **Thin circular layer**

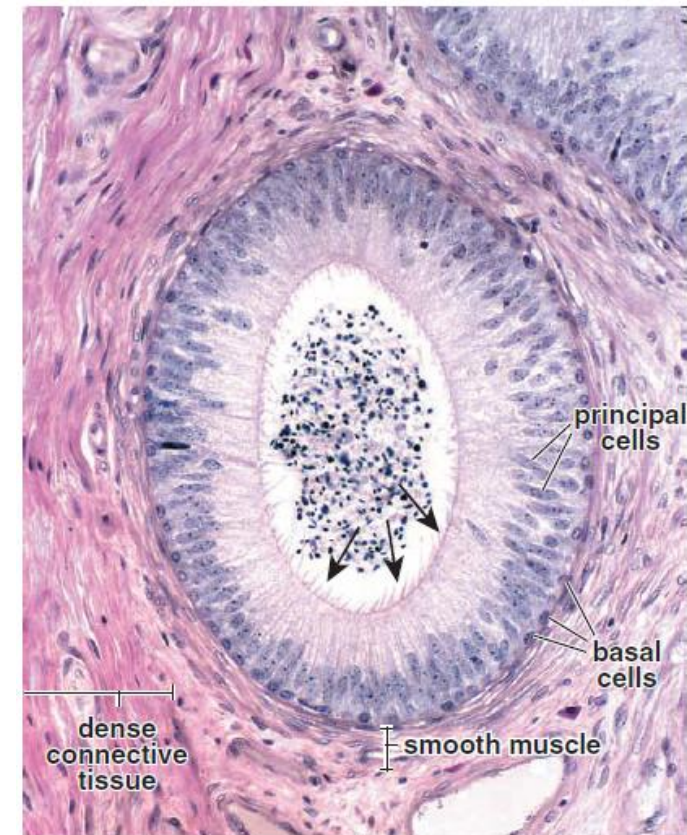
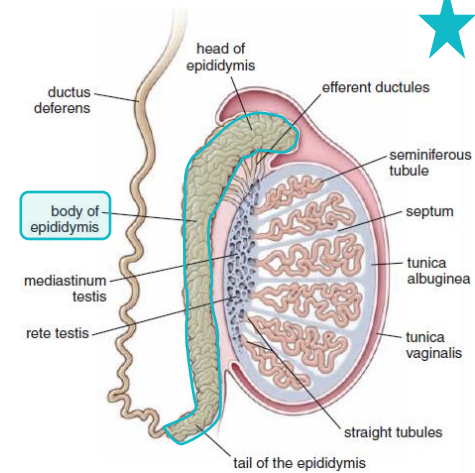
- *Head and most of body*
- Spontaneous, rhythmic peristaltic contractions
  - Move sperm along duct

- **Longitudinal layers**

- *Inner + outer layers added in tail*
- Fewer peristaltic contractions
  - **Principal reservoir of mature sperm**
- Intense contractions of all 3 layers in ejaculation
  - Sperm forced into ductus deferens

- Continuous with 3 layers of vas deferens

- **Lumen: smooth appearance**



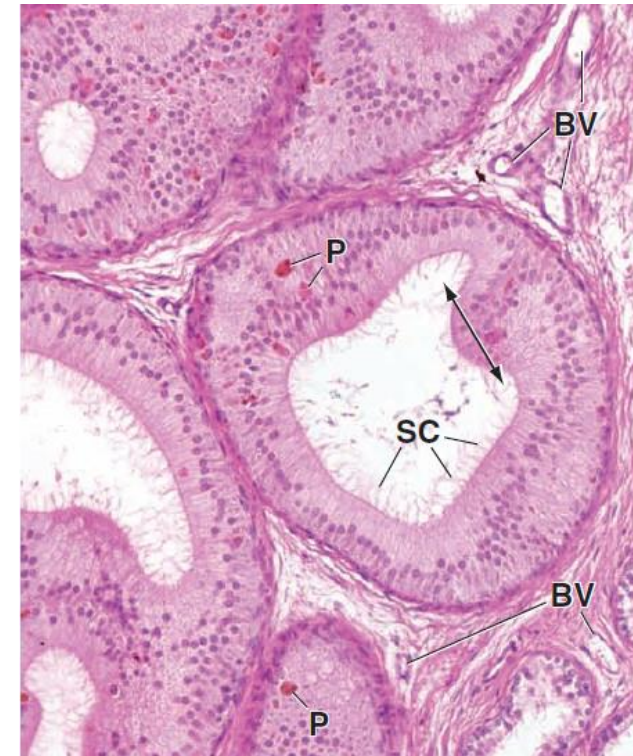
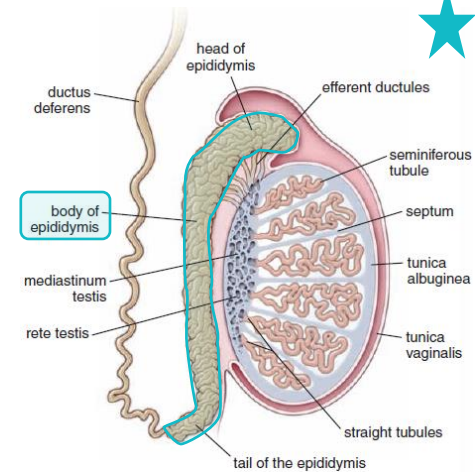




# Epididymis

## Functions:

- **Absorption (proximal epididymis)**
  - fluid not reabsorbed in efferent ductules
  - phagocytose residual bodies, degenerated sperm
- **Secretion (see previous slide)**
- **Androgen-dependent sperm maturation**
  - **Acquire motility and ability to fertilize oocyte**
  - **Decapacitation**
    - Surface-associated decapacitation factor modifies head of sperm
    - **Reversibly inhibits fertilizing ability of sperm**
    - Released during capacitation (in female reproductive tract)
- **Storage (distal epididymis)**
  - *Takes sperm ~12 days to course through duct*



**SC: stereocilia**

**P: Pigment**

**BV: Blood vessel**

# Ductus (Vas) Deferens

## • Epithelium

- **Pseudostratified columnar epithelium**
  - Secretory function (ampulla, ejaculatory duct)
- Thrown into deep longitudinal folds (fixation process)
  - Ampulla has taller, branched folds

## • Cells

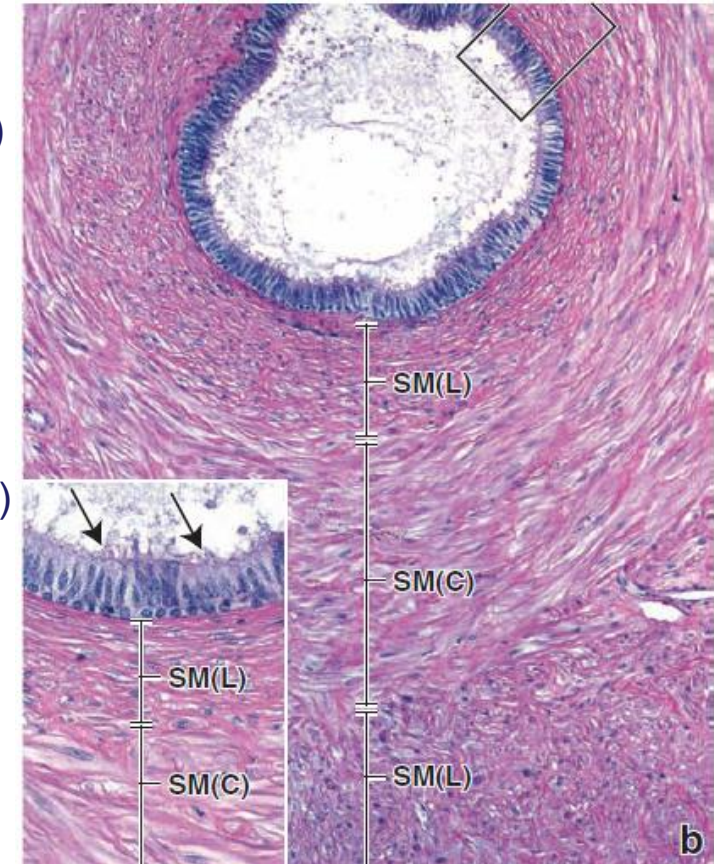
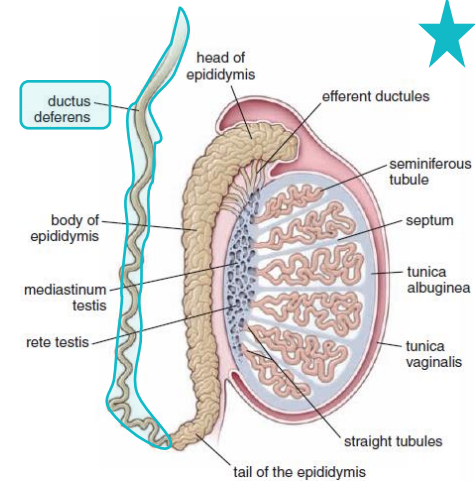
- **Tall principal cells** with stereocilia (*arrows*)
- **Round basal cells**

## • Smooth muscle

- **Longitudinal layers [SM(L)]**
  - Inner and outer layers
  - Thinner coat around ampulla
  - Disappears distally (around ejaculatory duct)
- **Circular layer [SM(C)]**

## • Direct continuation of epididymis

- **Transport sperm from epididymis to ejaculatory duct**



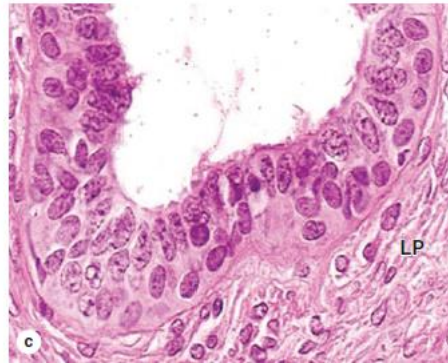
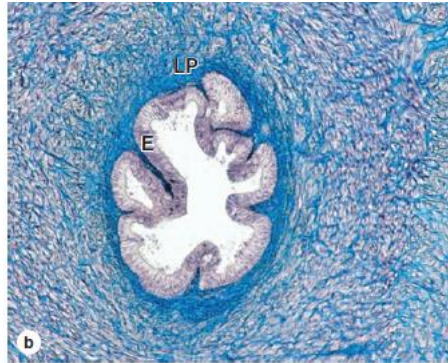
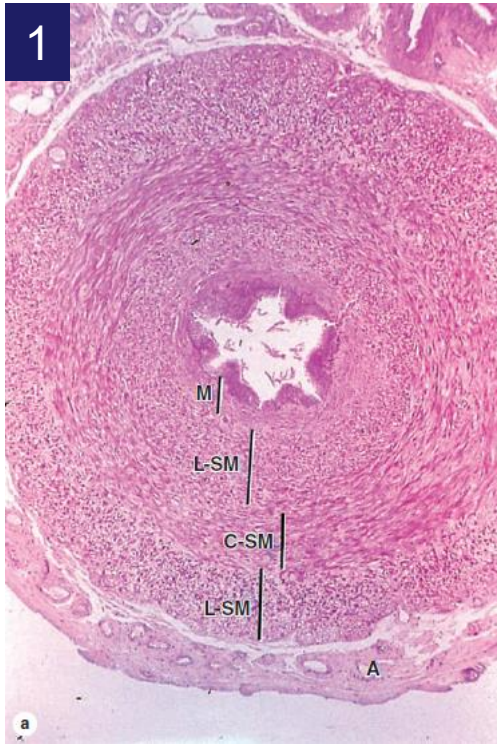


# Ductus (Vas Deferens)

## 1. Complete the labels

a. L-SM: \_\_\_\_\_

b. C-SM: \_\_\_\_\_



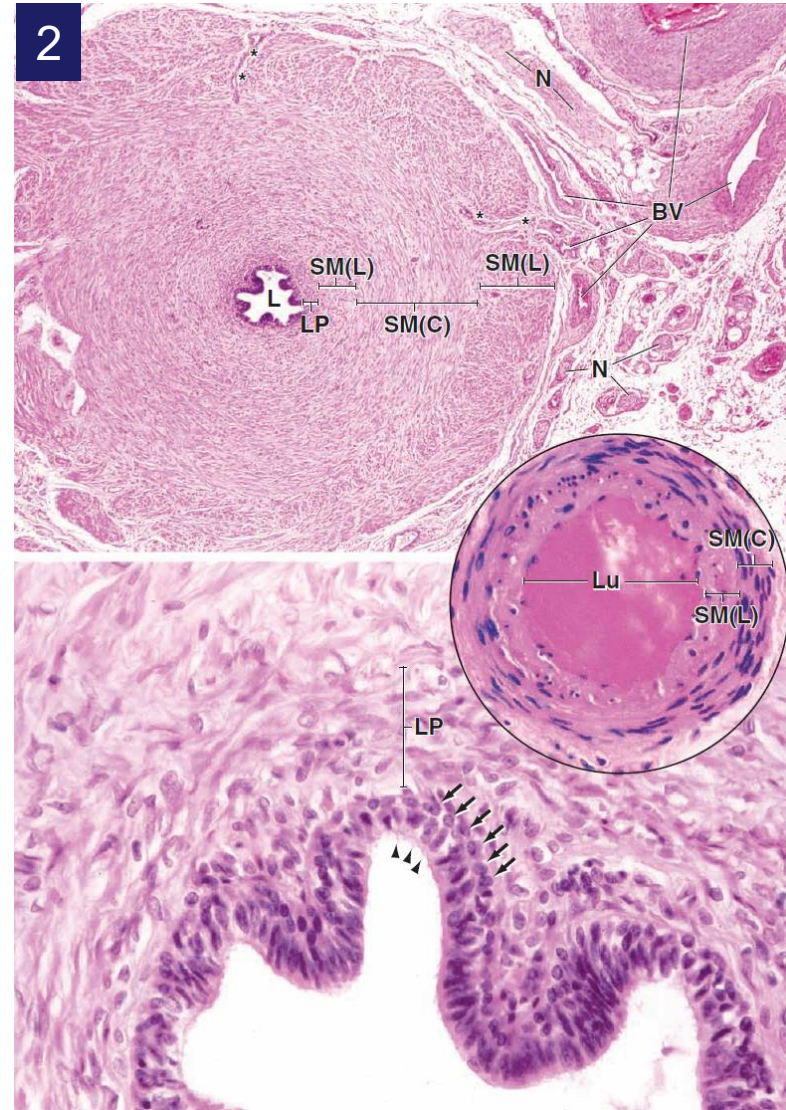
## 2. Complete the labels

a. SM(L): \_\_\_\_\_

b. SM(C): \_\_\_\_\_

c. Arrow: \_\_\_\_\_

d. Arrowhead: \_\_\_\_\_





# Penis

## • Functions:

- **Excretory:** terminal part of the urinary system (urination)
- **Reproduction:** emission of semen into the female genital tract

## • **Tunica albuginea**

- Dense fibroelastic layer encapsulating erectile bodies

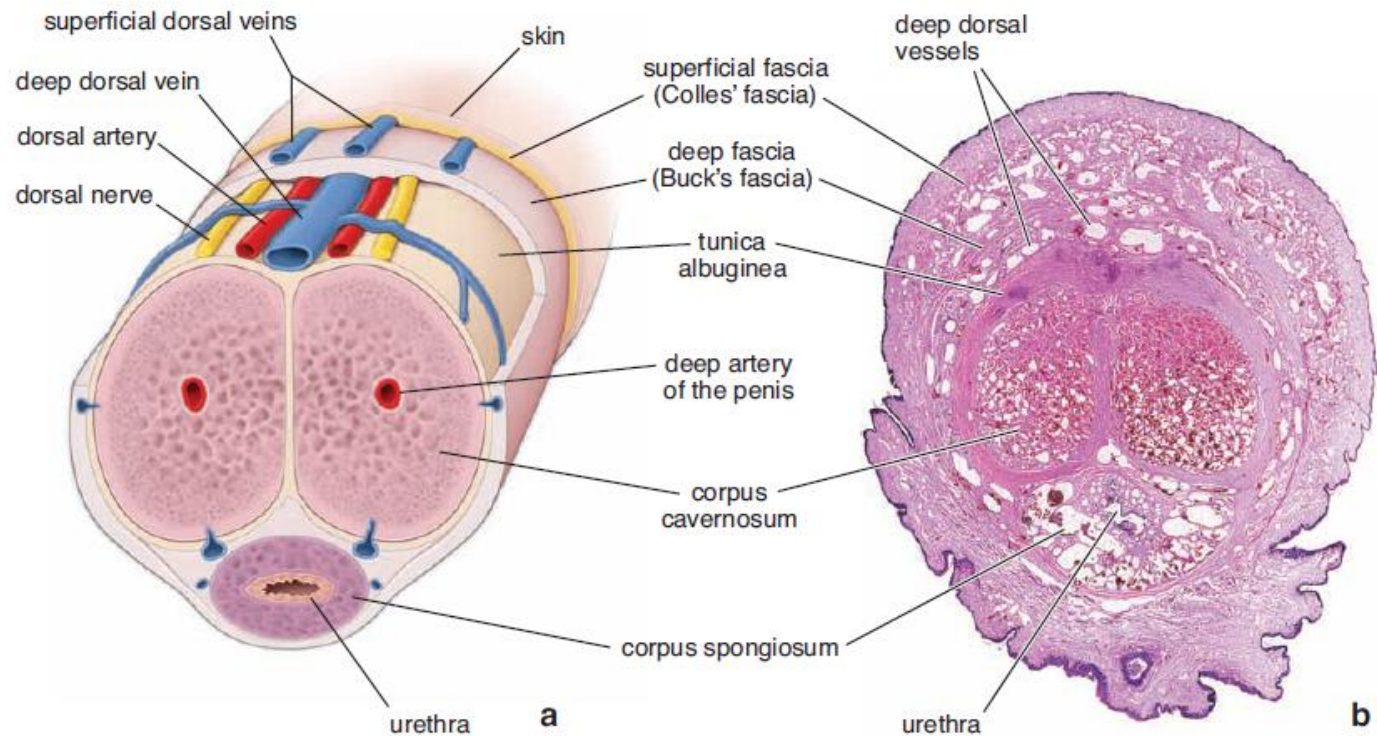
## • **Three (3) erectile bodies**

### ○ **Corpora cavernosa**

- 2 dorsal erectile bodies

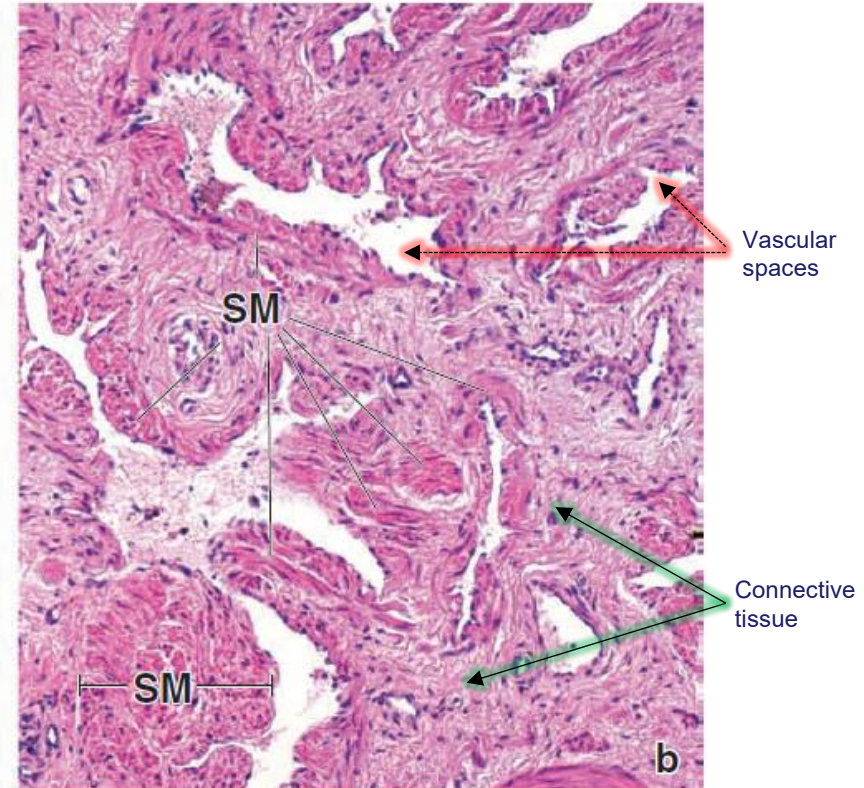
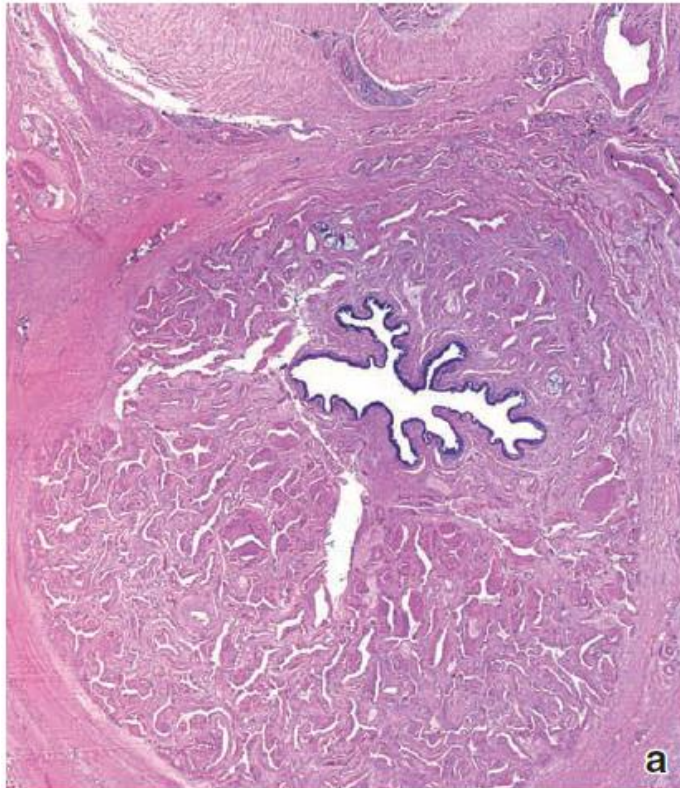
### ○ **Corpus spongiosum**

- 1 ventral body
- Contains the penile (spongy) urethra



# Penis

- Numerous, wide, irregular **vascular spaces**
- **Subendothelial cushions**: smooth muscle (SM) bundles surrounding vascular spaces
- **Connective tissue** with nerve endings and lymphatics
- Erectile tissues supplied by helicine arteries
  - Vascular spaces fill during erection





# Male Urethra

- 3 Histologic Regions

## 1. Prostatic

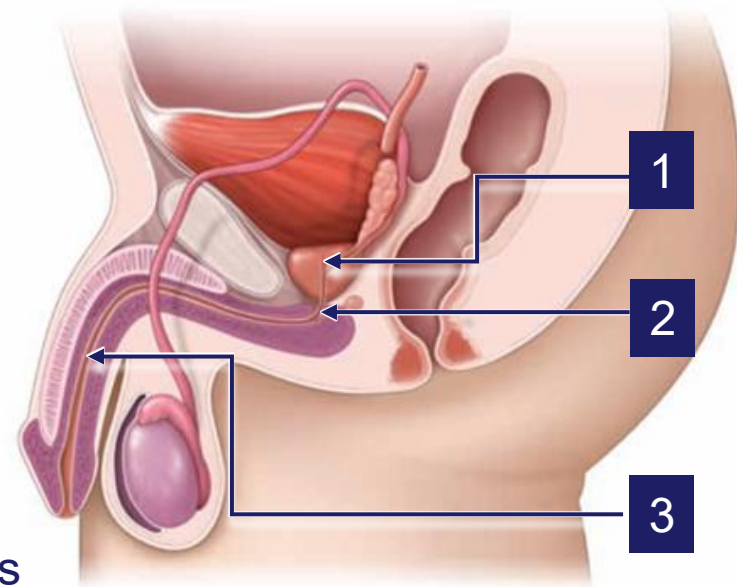
- ~3 to 4 cm from bladder neck
- Traverses through *prostate*
- Lined with *transitional epithelium*

## 2. Membranous

- ~1 cm from prostate apex to bulb of penis
- Lined with *stratified or pseudostratified columnar epithelium*
- Surrounded by external urethral sphincter (striated muscle)

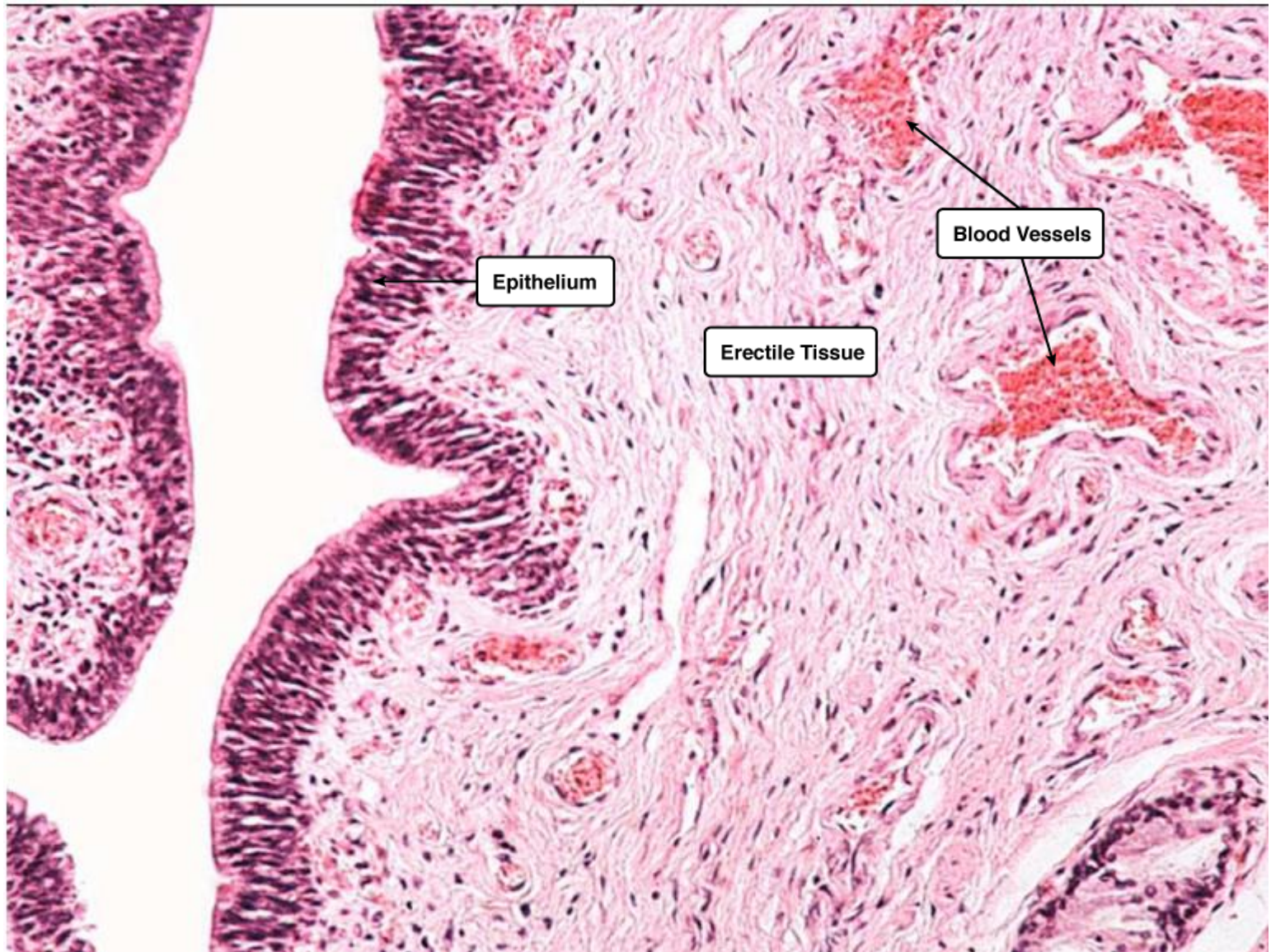
## 3. Penile (Spongy)

- ~15 cm through length of penis (*corpus spongiosum*)
- Lined with *pseudostratified columnar epithelium*
  - *Stratified squamous epithelium at urethral orifice*
- Receives the ducts of bulbourethral glands (Cowper's glands)
- Mucus secreting urethral glands (of Littre)





# Penile Urethra



[https://medcell.org/histology/male\\_reproductive\\_system\\_lab/urethra.php](https://medcell.org/histology/male_reproductive_system_lab/urethra.php)

# Semen Composition

Average ejaculate ~3mL

Alkaline secretion (pH 7.7) - neutralize acid environment (urethral + vagina)

## Contents:

- **Sperm Cells**

- Normal: up to 100 million sperm cells per milliliter
  - 20% morphologically abnormal
  - 25% immotile

- **Fluids – secretory products from**

- **Testes (2 – 5%)**
  - Testicular fluids not absorbed in straight tubules
- **Epididymis, Ductus deferens**
- **Seminal vesicles (65 – 75%)**
  - Nutrients (amino acids, citrates, fructose); prostaglandins
  - Provides protection for sperm cells
- **Prostate (25 – 30%)**
- **Bulbourethral glands (<1%)**

# Semen Analysis

| Parameter            | Reference Values         | Unit                         |
|----------------------|--------------------------|------------------------------|
| Volume               | 1.4-6.2                  | ml                           |
| Color                | Light cream/gray, opaque | Gross visual evaluation      |
| pH                   | 7.2-7.8                  |                              |
| Sperm Concentration  | $\geq 15$ million        | # sperm per ml               |
| Total Sperm Count    | $\geq 39$ million        | # sperms in total sample     |
| Total Sperm Motility | $\geq 42\%$              | #motile versus non motile    |
| Vitality/Viability   | $\geq 54\%$              | % live sperm in total sample |
| Morphology           | $\geq 4\%$ normal        | % of ideally shaped sperms   |
| WBC                  | $< 1$ million            | # WBC present per ml         |